Task	Tuned lens ρ	Logit lens ρ	Final acc
ARC-Easy	**0.577**	0.500	69.7%
ARC-Challenge	**0.547**	0.485	32.4%
LogiQA	**0.498**	0.277	21.4%
MNLI	0.395	**0.435**	40.4%
PiQA	**0.660**	0.620	76.1%
QNLI	**0.409**	−0.099	53.0%
QQP	**0.585**	−0.340	0.381 (F1)
RTE	0.156	**0.347**	60.0%
SciQ	**0.530**	0.505	91.9%
SST-2	**0.555**	0.292	64.7%
WinoGrande	0.517	**0.537**	63.9%

Table 2. Correlation between two measures of example difficulty, the iteration learned and prediction depth, across tasks. Prediction depth is measured using the tuned lens in the first column and the logit lens in the second.

prediction depth for every example. As a baseline, we also compute prediction depths using the logit lens. Finally, for each task, we compute the Spearman rank correlation between the iteration learned and the prediction depth across all examples.

Results. We present results in Table 2. We find a significant positive correlation between the iteration learned and the tuned lens prediction depth on all tasks we investigated. Additionally, the tuned lens prediction correlates better with iteration learned than its logit lens counterpart in 8 out of 11 tasks, sometimes dramatically so.

6. Discussion

In this paper, we introduced a new tool for transformer interpretability research, the *tuned lens*, which yields new qualitative as well as quantitative insights into the functioning of large language models. It is a drop-in replacement for the logit lens that makes it possible to elicit interpretable prediction trajectories from essentially any pretrained language model in use today. We gave several initial applications of the tuned lens, including detecting prompt injection attacks.

Finally, we introduced *causal basis extraction*, which identifies influential features in neural networks. We hope this technique will be generally useful for interpretability research in machine learning.

Limitations and future work. One limitation of our method is that it involves training a translator layer for each layer of the network, while the logit lens can be used on any pretrained model out-of-the-box. This training process, however, is quite fast: our code can train a full set of probes in under an hour on a single 8×A40 node, and further speedups are likely possible. We have also released tuned lens checkpoints for the most commonly used pre-

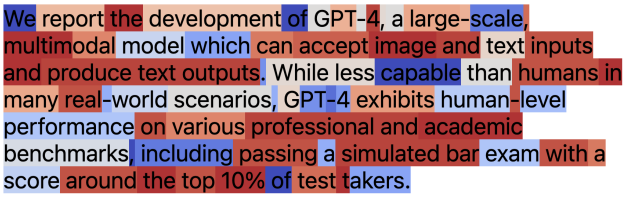

Figure 12. Token-level prediction depths for Pythia 12B computed on the abstract of OpenAI (2023). Warm colors have high prediction depth, while cool colors indicate low depth.

trained models as part of our `tuned-lens` library, which should eliminate this problem for most applications.

Causal basis extraction, as presented in this work, is computationally intensive, since it sequentially optimizes d_{model} causal basis vectors for each layer of the network. Future work could explore ways to make the algorithm more scalable. One possibility would be to optimize a whole k-dimensional subspace, instead of an individual direction, at each iteration.

Due to space and time limitations, we focused on language models in this work, but we think it's likely that our approach is also applicable to other modalities.

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